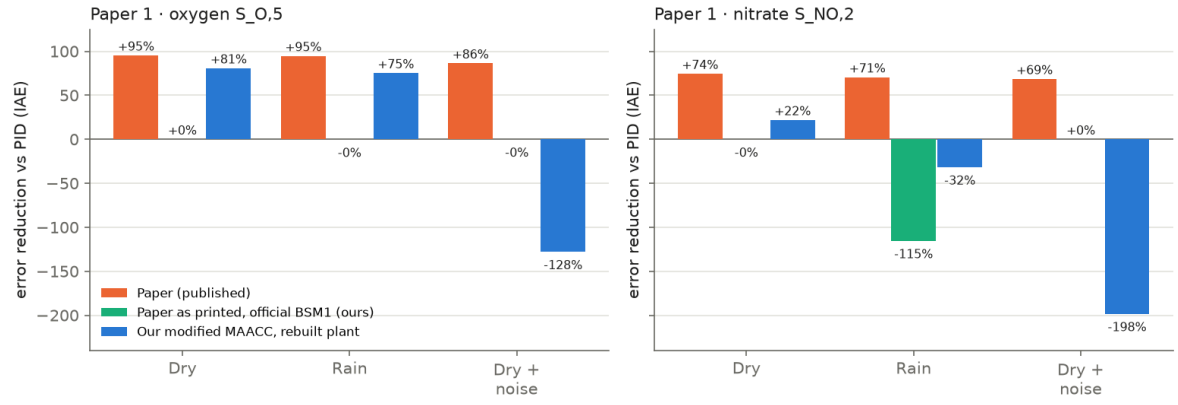
Base paper replication — results

Both papers run on the official BSM1 benchmark plant (bsm2-python) with the official influent · github.com/Amritha902/water_resource

	Paper 1 — MAACC (Wang et al., IEEE T-II 2026)	Paper 2 — floating DO (Du et al., Sci China 2023)
Claimed	Oxygen error −95%, nitrate −74% vs PID; >90% of 20 runs succeed	7.24% (dry) / 5.52% (rain) aeration energy saved vs PID
Plant check	Our PID within ~30% of the paper's PID row (dry)	PID & fuzzy within 1% of the paper (energy, effluent index)
Exactly as printed	MAACC = PID in all 180 runs; no initial weights reproduce Table II	DDPG-B saves 10.6% / 7.1% (3 seeds); one seed lands on the paper; ammonia over limit 56% vs 21% (PID)
Verdict	**Not reproducible from the paper as printed**	**Reproduced in direction; size depends on seed**

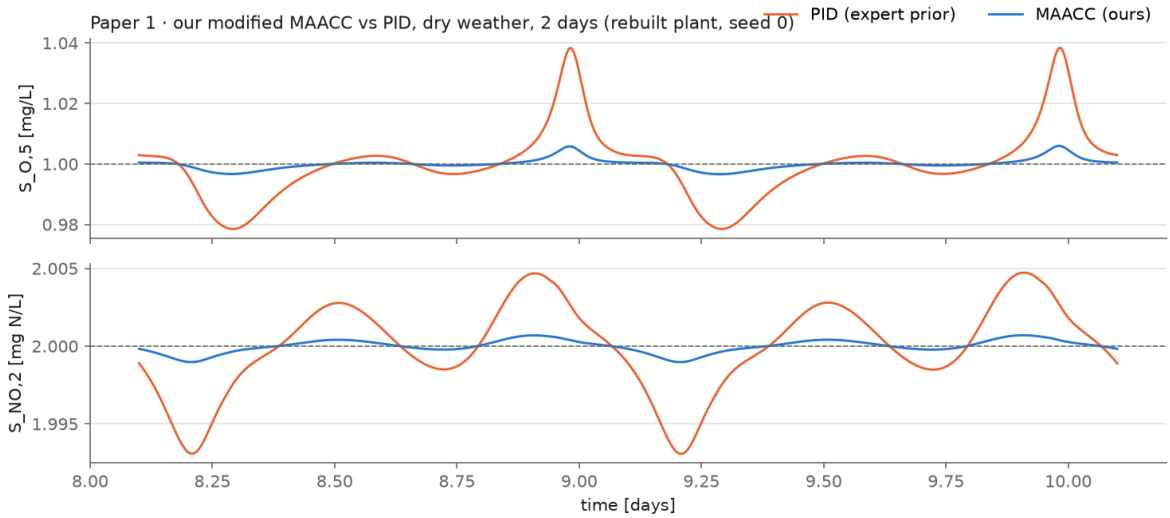
Paper 1 — error reduction vs PID



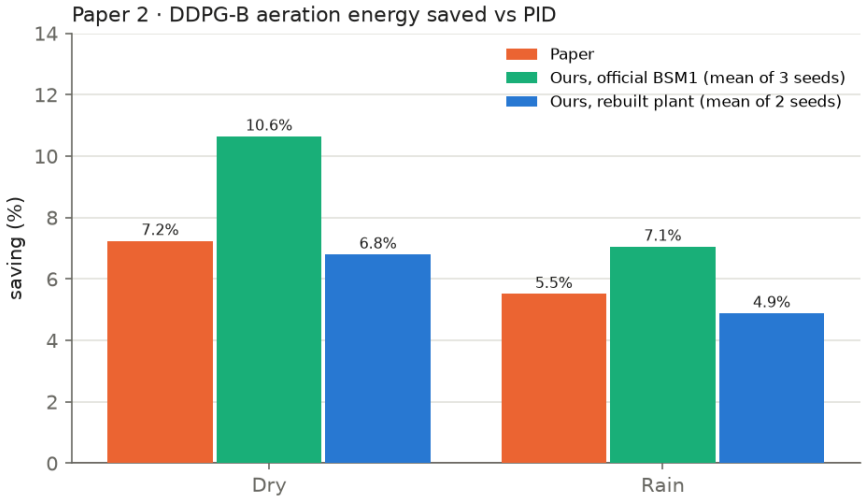
Mean of 20 runs per case. Up = MAACC better than PID. As printed, MAACC equals PID (0%) or does worse - the published gains are not reproduced.

Orange = published. Green = paper exactly as printed on the official plant (equals PID). Blue = our modified version (bigger steps, scaled inputs, safety fallback) on our rebuilt plant — not the paper's algorithm.

Paper 1 — our modified version tracking the set points



Paper 2 — DDPG-B energy saving vs PID



Exactly as published (1 seed): shuts the air almost off mid-training, ends at a 10.7% saving.
Storm: no official file available, so not in the exact runs.

Full summary of the runs

Every number below is from saved runs in results/ · details in docs/REPRODUCTION.md

What was done

Step	What	Runs
1	Official BSM1 plant (bsm2-python, port of the IWA benchmark code) + official dry and rain influent; rain file repaired (flows had '.' as thousands separator)	—
2	Paper 1 exactly as printed: eqs. (7)–(19), (27), Table I values, raw units; 3 readings of the critic update; dry, rain, dry + noise; 20 seeds	202
3	Paper 1: sweep of the only unstated setting (initial weight sizes), 12 settings × 2 critic readings	24
4	Paper 1, our modified version on our rebuilt plant: dry, rain, storm, noise; 20 seeds	300
5	Paper 2 on the official plant: PID, fuzzy, DDPG-A, DDPG-B (3 seeds each), dry and rain; plus DDPG-B exactly as published	17

Paper 1 — exact replication (official plant)

Case	IAE $S_{O,5}$ PID: paper / ours	IAE $S_{NO,2}$ PID: paper / ours	MAACC vs PID: paper	MAACC vs PID: as printed (ours)
Dry	0.0114 / 0.0082	0.0145 / 0.0095	−95% / −74%	0% / 0% · 0 of 20 runs better
Rain	0.0098 / 0.0077	0.0204 / 0.139*	−95% / −71%	≈0% / worse · 0 of 20
Dry + noise	0.0125 / 0.0245	0.0245 / 0.0160	−86% / −69%	0% / 0% · 0 of 20

* In the official rain event the PID drives Q_a to its 92 230 m³/d limit and nitrate falls to ≈0.6; the paper's PID does not, so its rain run differed. Initial-weight sweep: every setting either equals PID or diverges; the best (TD(0)) improves oxygen only (−68%). The learned term stays ~10^{-5} because the raw-unit utility is ~10^{-4}.

Paper 1 — our modified version (not the paper's algorithm)

Case	Oxygen vs PID (mean / median)	Nitrate vs PID (mean / median)	Runs beating PID

Dry	−81% / −88%	−22% / −84%	18 / 20
Rain	−75% / −88%	+32% / −83%	18 / 20
Storm (rebuilt influent)	−32% / −45%	+11% / +5%	4 / 20
Dry + noise	+128% (worse)	+198% (worse)	0 / 20

Changes: 10× step size and authority, normalised inputs, TD(0) critic, safety fallback that hands control back to the PID (without it 8 of 20 runs diverged), critic restart after repeated fallbacks.

Paper 2 — official plant, evaluation week

Weather	Method	AE paper / ours [kWh/d]	EQ paper / ours	S_NH paper / ours	Saving vs PID paper / ours
Dry	PID	3698 / 3728	6226 / 6196	2.34 / 2.81	—
Dry	DDPG-A (3 seeds)	3556 / 3653	6069 / 6319	2.65 / 3.40	3.8% / 2.0%
Dry	DDPG-B (3 seeds)	3430 / 3331	6073 / 6717	3.26 / 5.03	7.2% / 10.6%
Rain	PID	3666 / 3691	8220 / 8296	2.88 / 3.41	—
Rain	DDPG-A (3 seeds)	3584 / 3524	8147 / 8631	3.11 / 4.40	2.2% / 4.5%
Rain	DDPG-B (3 seeds)	3463 / 3430	8242 / 8703	3.65 / 4.72	5.5% / 7.1%

Fuzzy matches PID within 0.03% (paper also). Best seed, dry DDPG-B: AE 3513, DO 1.01, S_NH 3.31 vs paper 3430, 1.03, 3.26. Exactly as published (1 seed, dry): air nearly off mid-training, ends at 3328 kWh/d (10.7%), DO 0.51. Not in the paper: ammonia above 4 g N/m³ for 56% of the week with DDPG-B vs 21% with PID.

Corrected along the way

Paper 1's Table I (all hyperparameters) is an image and was first missed; the exact runs use it. "Paper 2 as published cannot learn" was true only on our rebuilt plant; corrected. Not covered exactly: storm weather (no official storm file reachable).

Two-minute talk for faculty

Reproducing two RL papers on the official BSM1 benchmark · about 2 minutes read aloud

Good morning. My work is on control of a wastewater treatment plant using reinforcement learning.

SET-UP

I ran both base papers on the official BSM1 benchmark plant, the simulator the papers use, with its official influent data. I followed each paper exactly as printed first, and only then changed anything.

PAPER 1

> Claim: −95% oxygen, −74% nitrate vs PID

Paper 1, by Wang and co-authors, puts two learning agents on top of a PID controller: one holds oxygen, one holds nitrate. They report errors about 95% lower than PID for oxygen and about 74% lower for nitrate.

> As printed: = PID in 180/180 runs

I implemented it exactly as printed, with every value from their Table I, and ran 20 runs per case, as they did. The plant checks out: my PID is within about 30% of their PID numbers. But the algorithm as printed does not learn. In all 180 runs its correction stays near zero, and the result equals the PID. I also tried twelve different starting weights, the one thing the paper leaves open; none reproduces their table. So their results cannot be obtained from the paper alone.

> Our variant: −81% oxygen (dry), 18/20 runs

To see what the idea can do, I built a modified version, with bigger steps, scaled inputs and a safety fallback to the PID. It cuts oxygen error by 81% in dry weather and beats PID in 18 of

20 runs. But that is my variant, not their algorithm, and under noise it is worse than PID.

PAPER 2

> PID ±1% · DDPG-B 10.6% / 7.1% (paper 7.2 / 5.5) · NH_4 over limit 56% vs 21%

Paper 2, by Du and co-authors, lets oxygen float to save aeration energy. They report 7.2% saving in dry weather and 5.5% in rain. On the official plant my PID and fuzzy comparators match their numbers within 1%. My DDPG agent saves 10.6% and 7.1% on average over three training seeds, and one seed lands almost exactly on their result. But the seeds spread widely. Run exactly as published, it shuts the air almost off during training, then recovers to a similar 10.7% saving. I also found what the paper does not report: with that saving, effluent ammonia is above its limit 56% of the week, against 21% under PID.

CONTRIBUTION

> −29 kWh/d at equal violations (rebuilt plant)

My contribution is a better agent for paper 2. It adds an influent forecast and a risk-aware critic, and it treats the limits as constraints. On our rebuilt plant, at the same violation rate, it uses about 29 kWh per day less than paper 2's agent.

CLOSE

In short: paper 2 reproduces in direction, paper 1 does not reproduce as printed, and every number I quoted comes from saved runs in the repository. Thank you.

Grey boxes are glance cues, not to be read aloud. Sources: github.com/Amritha902/water_resource — docs/REPRODUCTION.md (§2.0 paper 1 exact, §3.0 paper 2 official plant).